

U5-ATI REPORT

1. The Hybrid AI Recommendation Engine was designed to provide personalized recommendations by combining content-based filtering and collaborative filtering using PySpark RDDs. The system processes user-item ratings and item characteristics in a distributed environment, allowing large volumes of interaction data to be handled efficiently. Collaborative filtering identifies recommendations based on the preferences of similar users, while content-based filtering recommends items based on the characteristics of items previously preferred by the user. A hybrid score is generated by combining both approaches to improve recommendation quality. An intelligent recommendation agent monitors user feedback and adapts the recommendation process according to changing user interests. The implementation demonstrates how distributed processing and intelligent agents can be combined to build a scalable and adaptive recommendation system.
2. The Smart Disaster Detection System was developed to identify potential natural disasters by combining information from sensors, satellite sources, and social media. PySpark RDDs were used to distribute and process the collected data efficiently. The system calculates risk scores from each data source and applies a weighted data fusion technique to produce a final disaster risk score. Based on the calculated score, the system categorizes events into low-, medium-, or high-risk conditions. Intelligent agents are used to represent different monitoring functions, such as sensor monitoring, satellite analysis, and warning generation. The system demonstrates how distributed AI and multi-source data fusion can support faster disaster detection and early warning, helping authorities take appropriate action during potentially dangerous situations.
3. The AI-Based Fraud Detection Framework was implemented to identify suspicious financial transactions using distributed processing with PySpark RDDs. Transaction data containing user information, transaction amounts, locations, and transaction times was analyzed to establish normal transaction behavior for individual users. The system calculates user-specific average transaction values and compares new transactions against these patterns. Transactions involving unusually large amounts, unusual transaction times, or extremely high values receive higher fraud scores and can be classified as suspicious. An intelligent fraud detection agent monitors the classification results and generates alerts for potentially fraudulent transactions. The implementation demonstrates how RDD-based distributed processing and anomaly detection can be combined to analyze large-scale financial data and improve the speed and scalability of fraud detection..

4. The Autonomous Supply Chain Optimization System was developed to improve supply chain decision-making by analyzing demand, inventory, and transportation information using PySpark RDDs. The system distributes supply chain data and uses RDD transformations and aggregations to calculate total product demand, average demand, inventory requirements, and transportation costs. Based on predicted or observed demand, the system identifies products that require additional inventory and recommends reordering when stock levels are insufficient. Intelligent agents are used for different supply chain activities, including demand forecasting, inventory management, and logistics optimization. These agents support decentralized decision-making and allow different components of the supply chain to respond to changing conditions. The implementation demonstrates how distributed AI can help minimize inventory shortages, transportation costs, and operational inefficiencies.

5. The Personalized Learning Analytics Platform was developed to analyze student performance and learning behavior using PySpark RDDs. The system processes student scores and study hours in a distributed environment and calculates average performance for each learner. Based on the calculated scores, students are categorized into beginner, intermediate, or advanced learning levels. The system then generates personalized recommendations, such as basic learning materials for beginners, practice exercises for intermediate learners, and advanced projects for high-performing students. An intelligent learning agent analyzes each student's learning level and recommends suitable educational content. The implementation demonstrates how distributed data processing, AI-based learner analysis, and intelligent agents can be integrated to create an adaptive learning environment that can support a large number of students while providing personalized educational experiences.